# Math 1101: Calculus I Syllabus

**Instructor:** Cailan Li

**Place and Time:** 407 Mathematics Tu/Th 5:40PM-6:55PM

**TA:** Sayan Das `sayan@math.columbia.edu`

**TA help room hours:** 10-12 Wednesday 502 Milstein

**Course Content:**

- Functions and Models (Chapter 1)
- Limits and derivatives (Chapter 2)
- Differentiation rules (Chapter 3)
- Applications of Differentiation (Chapter 4)
- The integral (Chapter 5)

**Textbook:** James Stewart, Calculus: Early Transcendentals, 8th edition

**Prerequisites:** Understanding of Pre-Calculus. Knowledge of Calculus would also be helpful.

**Grading:** Your Grade will consist of

- Homework (20%)
- Midterm **March 14th in class** (30%)
- Final **May 13th 8-11AM in class** (50%)

**No make up exams will be given**. Welcome to the real world.

**Expectations:** I expect to pass some of you.

**Homework:** There will be weekly homework assignments due by **11:59 PM on Mondays** in the drop box on the 4th floor of the Math building. Late homeworks will be disposed. However, the lowest homework grade will be dropped. You are encouraged to discuss the homework with other students but you must write your solutions individually, in your own

words.

If you need help with your assignments, I highly enourage you to vist 502 Milstein help room in Barnard. You can drop by whenever it's open.

**Course Website:** Check https://math.columbia.edu/∼ccl/teaching/Calc1

**Office Hours:** Mon 13:00pm−15:00pm, 540 Mathematics Building

**Contact Information:** If you have questions relating to enrollment or administration please contact Alenia Reynoso at `reynoso@math.columbia.edu`. For matters relating to the course that doesn't include homework, you may email me directly at `ccl@math.columbia.edu`.

**Calendar:**

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| Week 1 | 1/28-2/1 | Review of Trig functions and Logarithms |
| Week 2 | 2/4-2/8 | Limits, Limit Laws, Squeeze Theorem |
| Week 3 | 2/11-2/15 | Continuity, IVT, Derviatives |
| Week 4 | 2/18-2/22 | Tangent Lines, Product and Quotient Rules |
| Week 5 | 2/25-3/1 | Chain Rule |
| Week 6 | 3/4-3/8 | Implicit Differentiation and Related Rates |
| Week 7 | 3/11-3/15 | Review and **Midterm** |
| Spring Break | 3/18-3/22 | Homework |
| Week 8 | 3/25-3/29 | Extreme Value Theorem and Mean Value Theorem |
| Week 9 | 4/1-4/5 | Second Derviative and L'Hopital's Rule |
| Week 10 | 4/8-4/12 | Optimization and Antiderivatives |
| Week 11 | 4/15-4/19 | Area, Riemann Sums, Definite Integrals |
| Week 12 | 4/22-4/26 | Fundamental Theorem of Calculus, Indefinite Integrals |
| Week 13 | 4/29-5/3 | Substitution Rule and Area between curves |
| Finals Week | 5/13-5/17 | **Final** |